Prasad Koviri

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Date of CV: June 2026

Personal Information

- **First Name**: Prasad, **Last Name**: Koviri
- **Gender**: Male; **Nationality**: Indian
- **Date (Place) of Birth**: 16 June 1998 (Vadanarsapuram, Andhra Pradesh, India).
- **Present Address**: 1300 South University Ave, APT 905, Ann Arbor, Michigan 48104-2677, USA.
- **Permanent Address**: H. No. 2-104, Beach Road, Vadanarsapuram, Rambilli, Anakapalle 531061, Andhra Pradesh, India.
- **Phone Numbers**: +1-734-216-8691 (Primary)
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Employment History

University of Michigan Ann Arbor, U.S.A.
Research Fellow, LSA Astronomy *March 2026 - Present*
Currently I am working with Prof. John Monnier and Prof. Steven Cundiff to demonstrate multi-heterodyne interferometry for astronomical applications utilizing EO comb technology.

The University of Electro-Communications Tokyo, Japan
Research Fellow at Shimizu laboratory, UEC Tokyo. *April 2025 - January 2026*
I worked with Prof. Ryosuke Shimizu to construct a quantum remote sensing testbed over a campus scale two-node fiber network including highly precise time-synchronization between remote time taggers utilizing White Rabbit precision time protocol.

Academic Curriculum

The University of Electro-Communications, @https://www.uec.ac.jp/ Tokyo, Japan
Ph.D. (Science); Advisor: Prof. Kaoru Minoshima *October 2021 - March 2025*
Dissertation: "Study of Quantum Entangled Ultrafast Two-Photon State by Time-Frequency Characterization Utilizing Optical Frequency Comb-Based Two-Color Asynchronous Optical Sampling"

University of Hyderabad, @https://uohyd.ac.in/ Hyderabad, India
Master of Science (5-year Integrated) Physics; CGPA: 9.16/10.00 *July 2016 - July 2021*
Thesis: "Polarization Transformation Properties of Dove Prism and Spin-Hall Effect of Light," Advisor: Prof. Nirmal K. Viswanathan

Jawahar Navodaya Vidyalaya Visakhapatnam, India
Senior School Examination (Mathematics, Physics, Chemistry, Computer Science), CBSE; Percentage: 92.60% *June 2013 - May 2015*

Jawahar Navodaya Vidyalaya Visakhapatnam, India
Secondary School Examination, CBSE; CGPA: 10/10 *July 2011 - April 2013*

Research Interests

During my Ph.D., I established an ultrafast time-resolved measurement technique leveraging optical frequency comb-based sampling technology allowing sub-100 fs level timing precision to probe the temporal structure of quantum-entangled ultrafast SPDC photon pairs, and demonstrated photonic quantum state engineering. In my Master's, I investigated the Spin-Hall effect of light at interfaces, such as a light beam propagating through a Dove prism, and reflecting at the air-prism interface. My research interests include, but are not limited to:

- Optical frequency comb and its applications, Time-Frequency metrology
- Ultrafast quantum optical technology, Quantum spectroscopy, Single-Photon detector technology
- Spin-Orbit interactions of light

Technical Skills

- **Laboratory Skills**: Hands-on experience with free-space and fiber optics, Mode-locked fiber lasers, Nonlinear crystals, Single-photon detectors (SNSPD, Si-APD), and Control of Time taggers with Python.
- **Programming Language**: MATLAB, Python, and C++
- **Software**: Origin, LATEX, MS-Office
- **Languages known**: Telugu (Native), English, Hindi, and Japanese (Introductory Level)

Research Publications

1. **P. Koviri**, T. Okita, R. Yabumoto, Y. Fujihashi, M. Yabuno, H. Terai, S. Miki, K. P. Nayak and R. Shimizu, "Quantum optical synthesis of high-dimensional ultrafast frequency-bin qudits", arXiV preprint (2026). http://arxiv.org/abs/2605.14314
1. **P. Koviri**, T. Kuwana, H. Komori, M. Ishizeki, T. Kato, A. Asahara, T. R. Schibli, R. Shimizu and K. Minoshima, "Time-frequency mode structure study of ultrafast two-photon quantum state with two-color dual-comb-based optical sampling", APL Photon. **10**(5), 051303 (2025). https://doi.org/10.1063/5.0253108
2. **P. Koviri**, H. Komori, H. Tian, M. Ishizeki, T. Kato, A. Asahara, R. Shimizu, T. R. Schibli and K. Minoshima, "Single-photon level ultrafast time-resolved measurement using two-color dual-comb-based asynchronous linear optical sampling", Appl. Phys. Express, **17**(2), 022001 (2024). https://doi.org/10.35848/1882-0786/ad2112

Conference Presentations and Proceedings

Quantum Innovation 2025, @Congres Square Grand Green Osaka — Osaka, Japan
Title (Poster): "Ultrafast biphoton time-frequency characterization with dual-comb-based optical sampling" — *July 29-31, 2025*

CLEO Pacific Rim 2024, @Songdo Convensia — Incheon, Korea
Title (Oral): "Time-Frequency Characterization of Ultrafast Quantum Entangled Biphotons Through Optical Frequency Comb Based Two-Color Asynchronous Optical Sampling" https://opg.optica.org/abstract.cfm?URI=CLEOPR-2024-Th2C_3 — *August 4-9, 2024*

SPIE Photonics West 2024, @The Moscone Center, San Francisco — California, USA
Title (Oral): "Ultrafast Time-resolved Single-photon Detection Using Two-color Comb based Asynchronous Optical Sampling for Quantum Applications" https://doi.org/10.1117/12.3002098 — *January 27 - February 1, 2024*

OSJ-JSAP Joint Symposia on Optics 2023, @Hokkaido University — Sapporo, Japan
Title (Oral): "Ultrafast Time-resolved Up-Conversion of Single-photons through Two-color Comb based Asynchronous Optical Sampling" — *November 27-29, 2023*

The 12th Advanced Lasers and Photon Sources, @PACIFICO Yokohama — Yokohama, Japan
Title (Oral): "Ultrafast Time-resolved Single-photon Detection Method Using Two-color Asynchronous Optical Sampling" — *April 18-21, 2023*

The 43rd Annual Meeting of The Laser Society of Japan, @WINC AICHI — Nagoya, Japan
Title (Oral): "Single-Photon Counting Asynchronous Optical Sampling using Dual-Wavelength Fiber Comb" — *January 18-20, 2023*

The 83rd JSAP Autumn meeting 2022, @Tohoku University — Sendai, Japan
Title (Oral): "Single-photon Level Detection of Ultrafast Cross-correlation Signal using Asynchronous Optical Sampling with Two-color Dual-comb System" — *September 20-23, 2022*

Student Conference on Optics and Photonics 2019, @Physical Research Laboratory — Ahmedabad, India
Title (Poster): "Spin Hall Effect of Light: Effect of Wavefront Curvature" — *September 24-26, 2019*

Workshops and Training Programs

IIST Astronomy and Astrophysics School (IAAS - 2020) — Trivandrum, India
Indian Institute of Space Science and Technology (IIST) — *December 18-28, 2020*

Physics Training and Talent Search (PTTS - 2018) — Surathkal, India
National Institute of Technology (NITK) — *May 21 - June 10, 2018*

National Initiative on Undergraduate Science (NIUS), Nurture and Exposure Camp 14.1 in Physics — Mumbai, India
Homi Babha Centre for Science Education (HBCSE), TIFR — *June 13-28, 2017*

National Science (VIJYOSHI) Camp - 2016 — Bengaluru, India
Indian Institute of Science (IISc) — *December 3-5,2016*

Innovation in Science Pursuit for Inspired Research (INSPIRE) Science Camp - 2013 — Visakhapatnam, India
Andhra University (AU) — *December 24-28, 2013*

Achievements

- The University of Electro-Communications student award for outstanding achievements in research activities for academic year 2024.
- MEXT Scholarship recipient, a doctoral fellowship awarded by the Japanese Government (October 2021 - September 2024).
- Qualified in Graduate Aptitude Test in Engineering (GATE - 2021, India) in Physics paper.
- Qualified in Jawahar Navodaya Vidyalaya Screening Test (JNVST) - 2008.

Hobbies

- Running, Playing volleyball, and cricket.
- Watch movies, and web series.

References

- **Prof. Kaoru Minoshima**
  Professor and Vice-President,
  The University of Electro-Communications,
  1-5-1 Chofugaoka, Chofu, Tokyo 182-8585, Japan.
  E-mail: k.minoshima@uec.ac.jp

- **Prof. Ryosuke Shimizu**
  Professor,
  The University of Electro-Communications,
  1-5-1 Chofugaoka, Chofu, Tokyo 182-8585, Japan.
  E-mail: r-simizu@uec.ac.jp